

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CE)

November 2013

CE302 System Programming

Date: 22.11.2013, Friday

Time: 10:00 a.m. To 1:00 p.m.

Maximum Marks: 70

Instructions: The question paper comprises of two sections.

1. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Define the following terms: [04]
1. Multiprogramming
2. binding
3. Semantic Gap
4. Program Relocation
- (b) What is an Operating System? Mention two functions performed by an Operating System. [03]
- Q - 2 (a) What is Memory Allocation? Explain its types. [06]

OR

- (a) Show the indirect triples' representation for the following program segment. [06]
 $Z = a + b * c + d * e - f$
 $Y = x + b * c$
- (b) What is input and output to an assembler? Describe the advantages of an Assembly Language over higher level language. [05]
- (c) Difference between one-pass and two-pass assembler. [03]
- Q - 3 (a) Explain naive approach of top down parsing. [06]
- (b) List the intermediate code generation techniques. Explain any one in brief. [04]
- (c) What is Editor? What are the functions of Editor? [04]

OR

- Q - 3 (a) Explain naive approach of bottom up parsing. [06]
- (b) What do you mean by ambiguity in grammar specification? Give an Example. [04]
- (c) Write a short note on "Forms of Text Editors". [04]

SECTION - II

- Q - 4 (a) Choose the correct answer. [04]
1. Which data structure in a compiler is used for managing information about variables and their attributes?
(A) abstract syntax tree (B) symbol table (C) semantic stack (D) Parse table
2. LEX is a
(A) lexical analyzer generator (B) A parser generator
(C) Code generator – generator (D) None of the above

3. In a compiler, keywords of a language are recognized during
 (A) parsing of the program (B) the code generation
 (C) the lexical analysis of the program (D) data flow analysis
4. Which one of the following languages over the alphabet $\{0, 1\}$ is described by the regular expression: $(0+1)^* 0 (0+1)^* 0 (0+1)^*$?
 (A) The set of all strings containing the substring 00
 (B) The set of all strings containing at most two 0's
 (C) The set of all strings containing at least two 0's
 (D) The set of all strings that begin and end with either 0 or 1.

(b) What is preprocessor? Explain functions of preprocessor. [03]

Q - 5 (a) What is meant by optimizing transformation? Explain any three with suitable example. [07]

OR

(a) With the help of an example, explain significance of static link and dynamic link. [07]

(b) What is swapping? What is paging? Explain significance of swapping and paging. [04]

(c) Justify: Problem oriented language makes execution gap larger. [03]

Q - 6 (a) Apply bottom of parsing using following grammar and reduce a string $+*42*39$ [07]

$E \rightarrow +TE \mid T$

$T \rightarrow *VT \mid V$

$V \rightarrow 0 \mid 1 \mid \dots \mid 9$

OR

(a) Apply top down parsing using following grammar and derive a string $+*42*39$ [07]

$E \rightarrow +TE \mid T$

$T \rightarrow *VT \mid V$

$V \rightarrow 0 \mid 1 \mid \dots \mid 9$

(b) Write difference between compiler and interpreter. [03]

OR

(b) Write difference between macro and subroutine [03]

(c) What is debugger? What are various facilities given by the debugger? [04]